

University of Moratuwa

Department of Electronic and Telecommunication
Engineering

Network Design Project



EN2150– Communication Network Engineering

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Contents

1	Introduction	2
1	Requirements and Problem Statement	2
2	Campus Backbone Network Design	3
1	Overview of the Campus Backbone	3
2	Selected Topology	4
3	ENTC Internal LAN Design	5
1	Overview of the ENTC Building Network	5
4	Active and Passive Components	7
1	Active Components	7
2	Passive Components	7
2.1	Single Mode Fibre	7
2.2	Multi Mode Fibre	8
2.3	Copper Cables	8
3	Device Specifications	8
5	IP Addressing Plan	10
1	IPv4 Addressing Scheme	10
2	IPv6 Addressing Scheme	11
6	Bill of Quantities (BOQ)	12
1	Active Components BOQ - Backbone	12
2	Passive Components BOQ - Backbone	12
7	Simulation in Cisco Packet Tracer	14
1	Backbone Network	14
2	ENTC LAN Network	15
3	Cisco Packet Tracer Files	15

Chapter 1

Introduction

1 Requirements and Problem Statement

The aim of this assignment is to design a network for the University of Moratuwa at two levels: the campus backbone and the internal network of the ENTC building. The backbone network should connect the main buildings of the university, while the ENTC network should represent how communication is organized inside one building.

The design should consider the fact that a backbone network is a long-term investment. Therefore, it should be planned in a way that is reliable, scalable, and cost-effective. The report must also include both active components, such as routers and switches, and passive components, such as cables, patch panels, racks, and outlets.

In addition, the ENTC building is used as a reference for studying a typical building LAN. Its internal network should be examined in terms of layout, cabling structure, and major networking devices. The proposed design should then be simulated using Cisco Packet Tracer.

In summary, the main problem is to create a practical and well-organized network design that connects the university buildings efficiently and also provides a clear internal network design for the ENTC building.

Chapter 2

Campus Backbone Network Design

1 Overview of the Campus Backbone

The campus backbone is designed using a hierarchical structure with two core switches at the center. These core switches form the main backbone of the university network and connect the major buildings together. Building-level switches are used to connect areas such as the Library, ENTC, L Block, and Sumanadasa to the core.

Inside each building, smaller departmental or area switches are connected to the main building switch. In this way, the backbone is responsible for communication between buildings, while the internal switches handle communication within each building.

Using two core switches improves the reliability of the backbone, because the network can continue operating even if one core link or device fails. This design is also easier to manage and expand in the future, which makes it suitable for a university campus network.

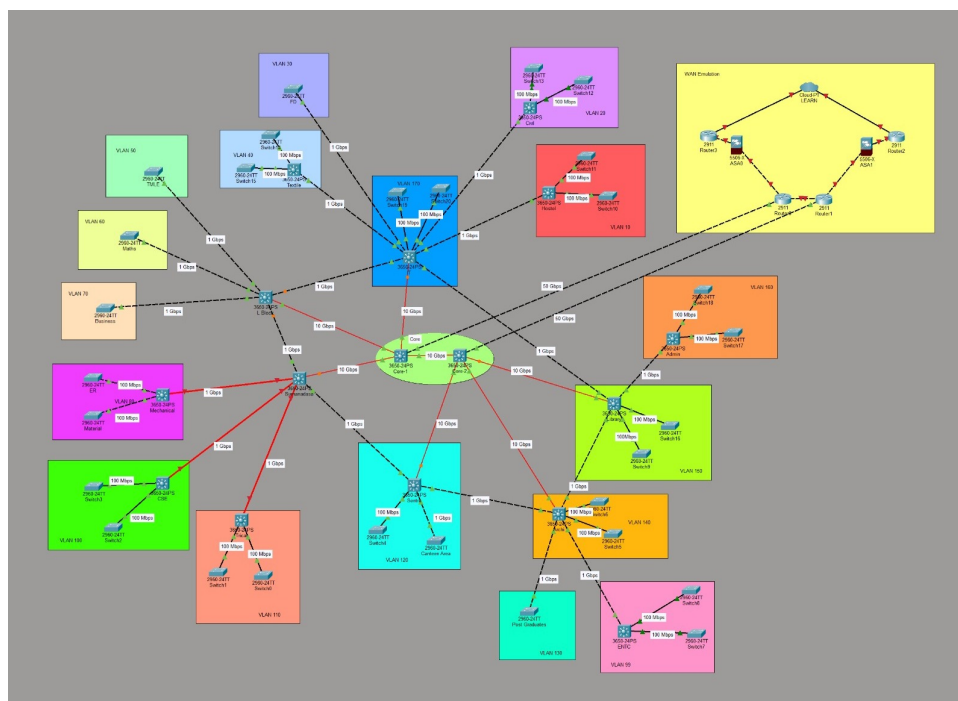


Figure 2.1: Backbone Network

2 Selected Topology

For the backbone, we selected a , centered around the core layer. In the diagram, the two core devices, Core 1 and Core 2, act as the central backbone nodes of the campus network. The main building-level switches are connected towards this core, either directly or through key aggregation points.

This approach is close to an extended star topology, but with an important improvement: instead of relying on only one central device, we used two core switches. This reduces the risk of the entire campus being disconnected due to a single core device failure.

- Centralized and easy to manage
- Better scalability
- Improved reliability
- Reasonable cost compared to full mesh

ENTC Internal LAN Design

1 Overview of the ENTC Building Network

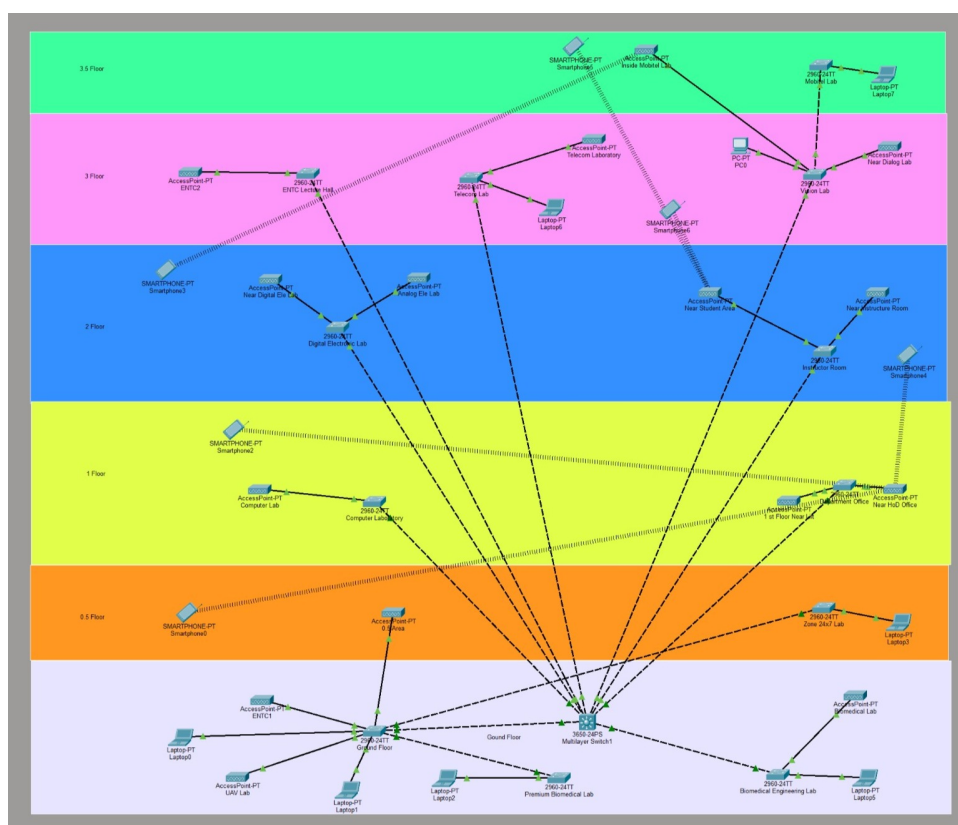


Figure 3.1: ENTC LAN Network

The ENTC LAN is designed as a multi-floor building network that connects laboratories, staff areas, and wireless users within the department. Since the building has several floors, the network is arranged in a structured way so that each floor can be connected efficiently while still being managed from a central point.

At the center of the LAN, a main multilayer switch is placed on the ground floor. This switch acts as the core device of the building network and connects the different floor-level switches together. From this central switch, links are extended to switches located in various labs and sections on different floors. In this way, the network follows

a hierarchical design, where one main device manages communication and several access switches serve end devices.

Each floor contains a number of access switches connected to PCs, laptops, and access points. This allows both wired and wireless connectivity throughout the building. The wired part of the network is mainly used for laboratories and fixed devices, while the wireless access points provide mobility for users such as students and staff using smartphones and laptops.

The design also shows that the network is distributed according to the physical arrangement of the building. Different labs and rooms on each floor are connected through nearby switches, which helps reduce cabling complexity and makes the network easier to organize and maintain. The uplinks from these floor switches to the central switch ensure that all parts of the building can communicate with each other and access shared services.

Overall, the ENTC LAN can be seen as a centrally managed, floor-wise structured network that supports both laboratory work and general user connectivity.

Chapter 4

Active and Passive Components

1 Active Components

- **Cisco 3650-24PS switches** were used for Core-1 ,Core-2 and department level because they are suitable for handling high traffic at the backbone level and can connect multiple building links efficiently. The selected core devices are multilayer switches, so they are able to perform both switching and routing functions. They were also selected because the backbone must support high traffic flow, and ordinary end-level access switches are not appropriate for this central role.

This has 24 ports.

- **Cisco Catalyst 2960-24TT** is a Layer 2 switch, mainly used for access -layer connectivity. It's suitable for connecting end-devices and smaller network segments. They support VLAN-based segmentation.

This switch also has 24 ports.

2 Passive Components

2.1 Single Mode Fibre

Single mode fibre was used between two cores, and the six nodes that are connected to the cores.

Single mode fibre was used between two cores,

- High bandwidth
- Low attenuation
- Longer distance
- Better future upgrade potential
- More reliable backbone performance

Single-mode fibre was selected for the links between the two cores and the six main nodes because these are critical backbone connections that require high capacity, low loss, and good long-term scalability.

2.2 Multi Mode Fibre

Multimode fibre was used in connecting from Sumanadase to Mechanical, CSE, Electrical Departments. Multimode fibre was used between Sumanadasa and its three connected nodes because these links are shorter local distribution links rather than major backbone connections. It provides sufficient bandwidth for this purpose while being more cost-effective and than single-mode fibre.

2.3 Copper Cables

Copper straight through cables are used to connect access points and switches in LANs (example: ENTC LAN).

3 Device Specifications

- **Cisco Catalyst 2960-24TT**

- Switch type: Fixed-configuration Layer 2 Ethernet switch
- Total ports: 26
- Access ports: $24 \times 10/100$ Fast Ethernet ports
- Uplink ports: $2 \times 10/100/1000$ Gigabit Ethernet uplink ports
- Forwarding bandwidth: 16 Gbps
- Maximum forwarding rate: 6.5 Mpps
- DRAM: 64 MB
- Flash memory: 32 MB
- MAC address table: 8,000 entries
- VLAN IDs supported: up to 255 active VLANs
- Frame size: up to 9000 bytes jumbo frame support
- Software feature set: LAN Lite / LAN Base depending on model variant

- **Cisco Catalyst 3650-24PS**

- Switch type: Multilayer switch (Layer 2 and Layer 3)
- Access ports: $24 \times 10/100/1000$ Gigabit Ethernet PoE+ ports
- PoE standard: IEEE 802.3at PoE+
- PoE power per port: up to 30 W
- Uplink options: fixed uplink model, commonly $4 \times 1\text{G}$ SFP uplinks depending on SKU
- Stack bandwidth: 160 Gbps with StackWise-160
- Switching capacity: up to 88 Gbps
- Forwarding rate: up to 65.48 Mpps
- DRAM: 4 GB
- Flash memory: 2 GB

- IPv4 and IPv6 support: yes
- Routing support: static routing, RIP, OSPF, EIGRP, BGP depending on license level
- Power supplies: modular, with redundant power supply support

Chapter 5

IP Addressing Plan

1 IPv4 Addressing Scheme

Table 5.1: Subnet allocation for each VLAN

VLAN	Users	Subnet	CIDR	Mask
140	3000	10.0.0.0	/20	255.255.240.0
170	2500	10.0.16.0	/20	255.255.240.0
10	2000	10.0.32.0	/21	255.255.248.0
80	1250	10.0.40.0	/21	255.255.248.0
100	1000	10.0.48.0	/22	255.255.252.0
150	1000	10.0.52.0	/22	255.255.252.0
70	800	10.0.56.0	/22	255.255.252.0
20	600	10.0.60.0	/22	255.255.252.0
40	600	10.0.64.0	/22	255.255.252.0
120	600	10.0.68.0	/22	255.255.252.0
99	600	10.0.72.0	/22	255.255.252.0
50	500	10.0.76.0	/23	255.255.254.0
110	500	10.0.78.0	/23	255.255.254.0
30	400	10.0.80.0	/23	255.255.254.0
160	100	10.0.82.0	/25	255.255.255.128
60	200	10.0.83.0	/24	255.255.255.0
130	200	10.0.84.0	/24	255.255.255.0

2 IPv6 Addressing Scheme

Table 5.2: IPv6 subnet allocation per VLAN

VLAN	IPv6 Subnet	Prefix	Assignment
10	2001:db8:0:a::/64	/64	SLAAC
20	2001:db8:0:14::/64	/64	SLAAC
30	2001:db8:0:1e::/64	/64	SLAAC
40	2001:db8:0:28::/64	/64	SLAAC
50	2001:db8:0:32::/64	/64	SLAAC
60	2001:db8:0:3c::/64	/64	SLAAC
70	2001:db8:0:46::/64	/64	SLAAC
80	2001:db8:0:50::/64	/64	SLAAC
99	2001:db8:0:63::/64	/64	DHCPv6
100	2001:db8:0:64::/64	/64	DHCPv6
110	2001:db8:0:6e::/64	/64	SLAAC
120	2001:db8:0:78::/64	/64	SLAAC
130	2001:db8:0:82::/64	/64	SLAAC
140	2001:db8:0:8c::/64	/64	SLAAC
150	2001:db8:0:96::/64	/64	DHCPv6
160	2001:db8:0:a0::/64	/64	DHCPv6
170	2001:db8:0:aa::/64	/64	SLAAC

Chapter 6

Bill of Quantities (BOQ)

1 Active Components BOQ - Backbone

Table 6.1: Bill of Quantities for Active Components

Device Type / Model	Quantity
Cisco 3650-24PS (Multilayer Switch)	17
Cisco 2911 Router	4
Cisco ASA 5506-X Firewall	2
Cloud-PT (WAN Emulator)	1

2 Passive Components BOQ - Backbone

Table 6.2: Single-mode Fibre Links

No.	Link	Cable Type	Length (m)
1	Core 1 – Core 2	Single-mode fibre	5
2	CITES – IT Faculty	Single-mode fibre	100
3	CITES – Sumanadase	Single-mode fibre	200
4	CITES – Library	Single-mode fibre	270
5	CITES – Central Court	Single-mode fibre	160
6	CITES – Archi	Single-mode fibre	110
7	CITES – L Block	Single-mode fibre	220
Total			1065

Table 6.3: Multimode Fibre Links

No.	Link	Cable Type	Length (m)
1	Sumanadasa – Trical	Multimode fibre	10
2	Sumanadasa – CSE	Multimode fibre	20
3	Sumanadasa – Mechanical	Multimode fibre	315
Total			345

Table 6.4: Copper Crossover Cable Links

No.	Link	Cable Type	Length (m)
1	L Block – IT	Copper crossover	80
2	IT – Library	Copper crossover	300
3	Archi – Central Court	Copper crossover	150
4	Central Court – Sumanadase	Copper crossover	10
5	Sumanadase – L Block	Copper crossover	230
6	Library – Archi	Copper crossover	50
7	IT – Textile	Copper crossover	230
8	IT – FD	Copper crossover	300
9	IT – Civil	Copper crossover	190
10	IT – Hostel	Copper crossover	200
11	Archi – ENTC	Copper crossover	50
12	Archi – Post Graduate	Copper crossover	80
13	Library – Admin	Copper crossover	100
14	Mechanical – Material	Copper crossover	270
15	Mechanical – ERE	Copper crossover	260
16	L Block – Business	Copper crossover	10
17	L Block – Maths	Copper crossover	20
18	L Block – TMLE	Copper crossover	110
19	Sentra – Canteen	Copper crossover	75
20	CITES Router connection 1	Copper crossover	5
21	CITES Router connection 2	Copper crossover	5
Total			2725

Table 6.5: Bill of Quantities for Passive Cabling Components

Item No.	Description	Unit	Quantity (m)
1	Single-mode fibre cable (Backbone)	m	1065
2	Multimode fibre cable (Department links)	m	345
3	Copper crossover cable (Interconnections)	m	2725
Total Cable Length			4135

Chapter 7

Simulation in Cisco Packet Tracer

1 Backbone Network

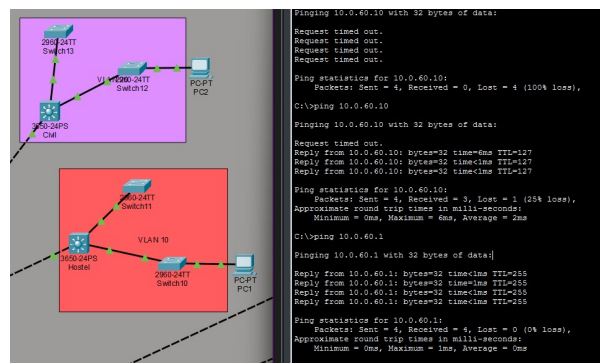


Figure 7.1: Simulation of Backbone

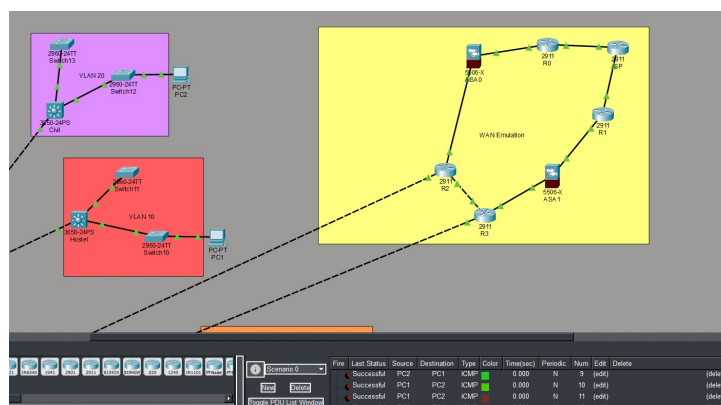


Figure 7.2: Simulation of Backbone

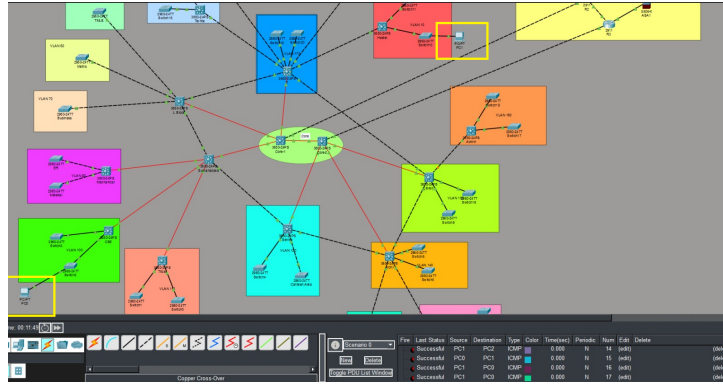
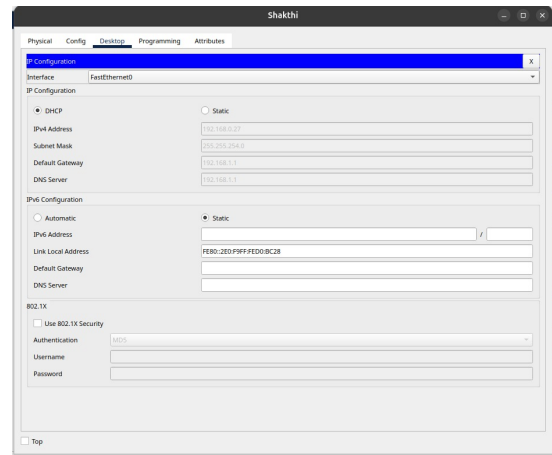
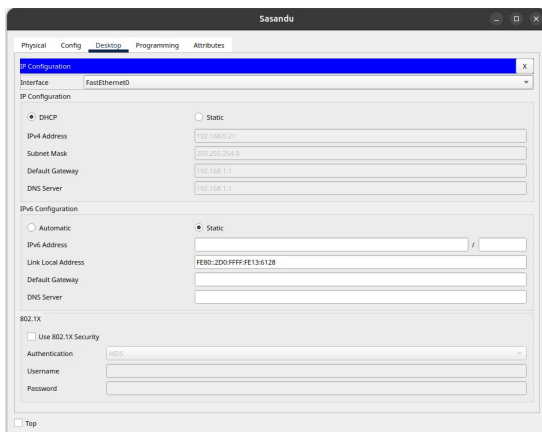


Figure 7.3: Simulation of Backbone



2 ENTC LAN Network

3 Cisco Packet Tracer Files

The Cisco Packet Tracer simulation files related to this network design are provided through Google Drive links for reference.

- PKT Files Link: [Open Packet Tracer File](#)

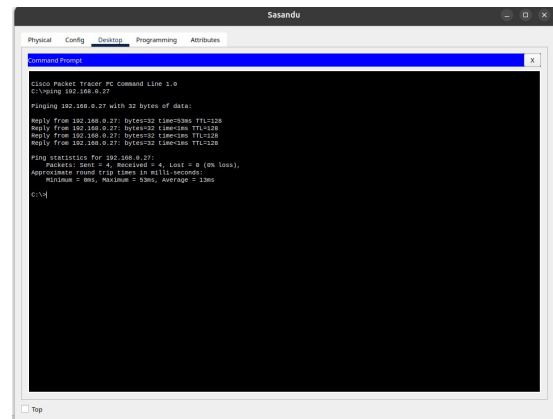
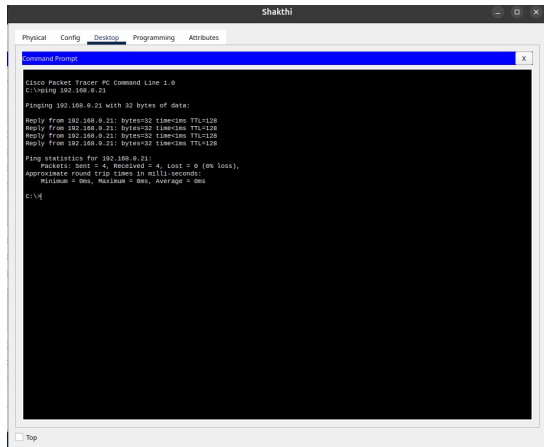


Figure 7.4: Simulation of ENTC LAN

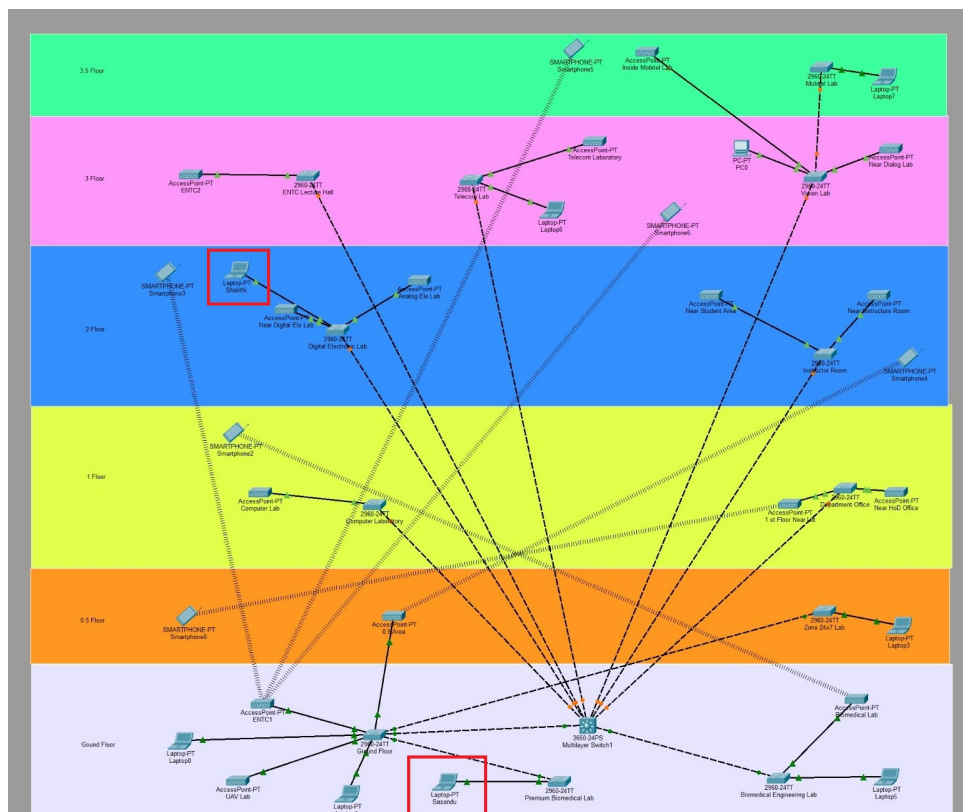


Figure 7.5: These two devices were used for simulation